

Theorem 1. *For every natural number $n \in \mathbb{N}$, there exists a prime number p such that $n \leq p$. Consequently, there are infinitely many prime numbers.*

Proof. Let $n \in \mathbb{N}$ be given. Consider the natural number $n! + 1$. Since $n! \geq 1$, we have $n! + 1 \geq 2$, so $n! + 1 \neq 1$.

Let p be the smallest prime factor of $n! + 1$. Since $n! + 1 \neq 1$, such a prime factor exists, and by definition p is prime.

We now claim that $n \leq p$. Suppose, for the sake of contradiction, that $p < n$. Since p is a positive prime with $p \leq n$, p divides the factorial $n!$ (that is, $p \mid n!$). Furthermore, by definition of p , we have $p \mid (n! + 1)$.

Since p divides both $n!$ and $n! + 1$, p must also divide their difference:

$$p \mid ((n! + 1) - n!) \implies p \mid 1.$$

However, no prime number divides 1, yielding a contradiction.

Hence, our assumption that $p < n$ must be false, so $n \leq p$. Thus, p is a prime number satisfying $n \leq p$, as required. $\square$